

LiDAR Bare-Earth DEM

Quality Control Memorandum

Site: SFWMD structure S-151, L-67A / Miami Canal intersection, Water Conservation Area 3, Broward / Miami-Dade county line, Florida

Source: USGS 3DEP, FL_Southeast_2018_D18_SUPPLEMENTAL, tile e1557n0456, acquired 2018, published 2019-10-15

CRS: EPSG:6350 — NAD83(2011) / Conus Albers, **metres**

Vertical: NAVD88, **Geoid12B** (declared in the LAS header)

Date: 2026-08-11

EXECUTIVE SUMMARY

A bare-earth DEM was produced from raw returns for the S-151 structure at the L-67A / Miami Canal intersection, Water Conservation Area 3, from the only 3DEP tile that covers the site. The controlling condition here is not point density but **ground** coverage: the tile carries 16.89 pts/m² overall yet only 1.26 pts/m² of ground, because 41.10% of it returns signal that never reaches the surface through sawgrass and a further 6.07% returns nothing at all over open water. A measured cell-size sweep set the working resolution at 3.0 m; at 1 m, 47.04% of cells would hold no ground observation at all.

Across the marsh — 94.6% of the tile — this surface agrees with the vendor's delivered ground classification to **0.081 m RMSE**. That figure is agreement between two classifications of the same point cloud, not accuracy: no external control exists on this tile, and an ASPRS Non-Vegetated Vertical Accuracy figure cannot be computed from it.

One limitation is structural and is stated rather than worked around. The L-67A levee crown is 6.0–8.5 m wide at its top, narrower than the smallest structuring element SMRF can form at this cell size, so the delivered surface truncates it by a median 0.911 m. The two requirements — ground coverage in the marsh and resolution of the crown — are irreconcilable within a single parameter set, and that conflict is a property of the site rather than a tuning failure. A finer cell recovers the crown at the cost of a 15× increase in marsh cells sitting above the vendor surface, which was measured rather than assumed.

Portfolio deliverable — bare-earth DEM processed from raw returns and compared against the vendor's delivered ground classification. Agreement, not accuracy: no external vertical control exists on this tile.

1. Scope, and what this document does not claim

This memo reports a bare-earth DEM produced from raw returns, and its **agreement** with the vendor's delivered ground classification.

It is not an accuracy assessment. No external vertical control was used: no NGS marks were sought on this tile, and USGS published no vertical accuracy report for this collection comparable to the sealed survey documents available for other projects. Every difference statistic below compares two independent classifications of the *same* point cloud. Both could be biased identically and nothing here would reveal it.

A Non-Vegetated Vertical Accuracy figure in the sense of the **ASPRS Positional Accuracy Standards for Digital Geospatial Data** cannot be computed from this tile alone; it would require new field survey. The agreement statistics in §6 are reported as agreement, and should not be quoted as RMSEz.

2. Why this site

The site was selected to invert, rather than repeat, the conditions of a prior desert project. Figures in the left-hand column are **[cited: lidar-portfolio CLAUDE.md]** — they were derived in that project and are quoted here, not re-derived from this tile. Right-hand figures are measured from this tile.

	prior site (Sonoran desert) [cited: lidar-portfolio CLAUDE.md]	this site
horizontal units	International feet	metres
geoid	12A, undeclared in header	12B, declared
ground share of returns	70.9%	7.4%
all-return density	3.74 pts/m ²	16.89 pts/m²
ground density	~2.6 pts/m ²	1.26 pts/m²
relief across tile	50 m	2.81 m (p1–p99)

Denser data, far less usable ground, and 2.81 m of total relief. Aggregate point density satisfies **USGS LiDAR Base Specification** QL1 thresholds, but aggregate density is the wrong statistic here: the operative constraint is *ground* return density at 1.26 pts/m², which is what drives every parameter in §4 and §5.

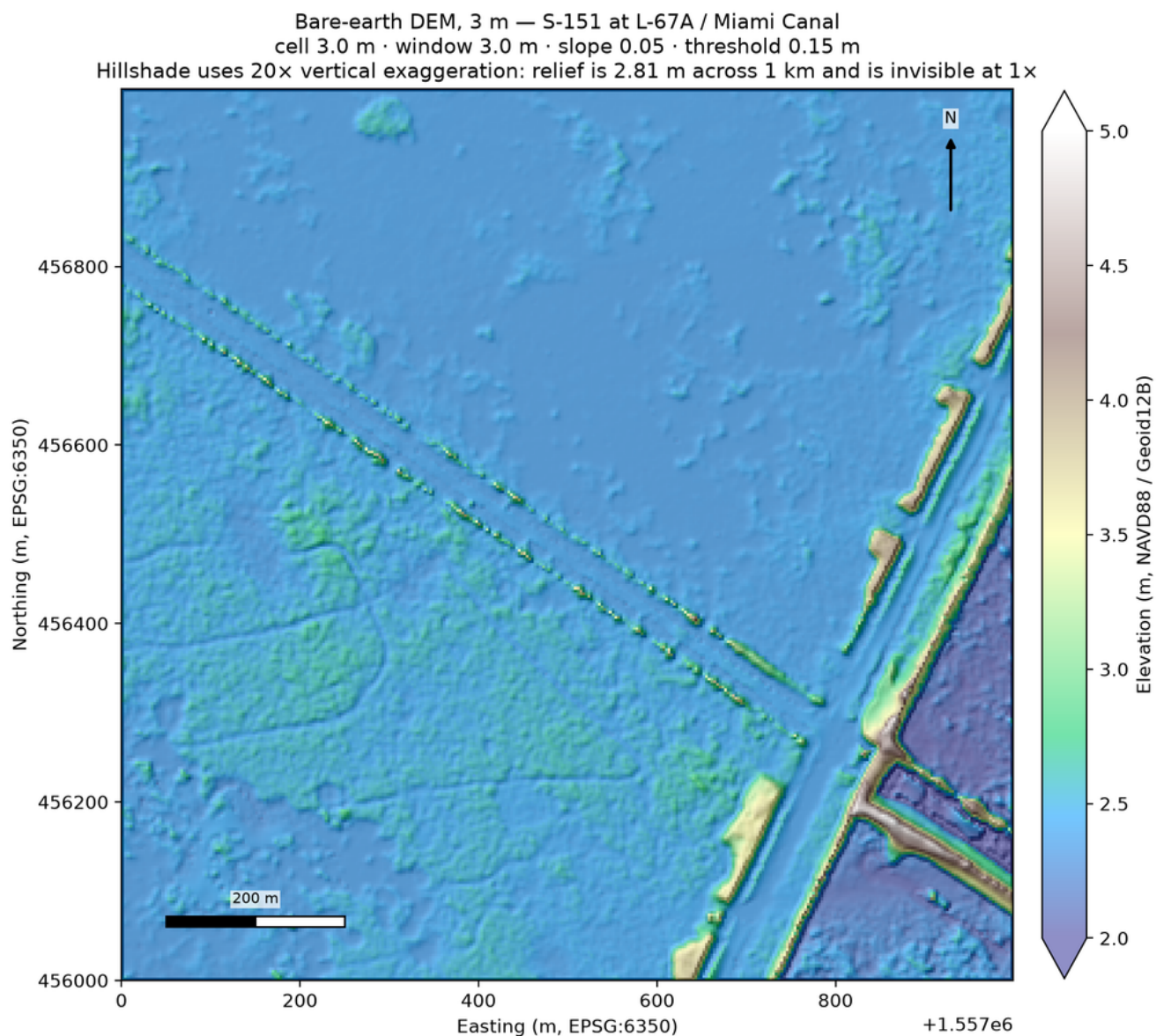


Figure 1. The delivered bare-earth surface, 3 m. Hillshade uses 20× vertical exaggeration because relief is 2.81 m across 1 km and is invisible at 1×. The diagonal is the L-67A levee; the canal crosses it at S-151.

3. Coverage is the binding constraint

Aggregate point density satisfies **USGS LiDAR Base Specification** QL1 thresholds. Aggregate density is nonetheless the wrong statistic for this site, and using it would lead to a resolution the data cannot support.

Per-cell counts were computed with true binning (`binmode`), not a radius search — the default radius method inflates apparent density several-fold and would badly mislead at 1.26 pts/m². Classifying every 1 m cell by what it actually received:

cause	share of tile
ground present	52.9%
returns present, none reaching ground — sawgrass	41.0%
no returns at all — open water	6.1%

These are two different failures and must not be reported as one number. Vegetation blocking is a classification problem: the returns exist and a filter can be asked to find ground among them. Open water is irreducible absence — the pulse is absorbed and nothing comes back, so no processing choice recovers it. 95.7% of the dropout area lies in clusters larger than 100 m², which is to say it is the canals rather than scattered noise.

The practical consequence appears in §4: the vegetation gap closes as cells grow, and the water floor does not.

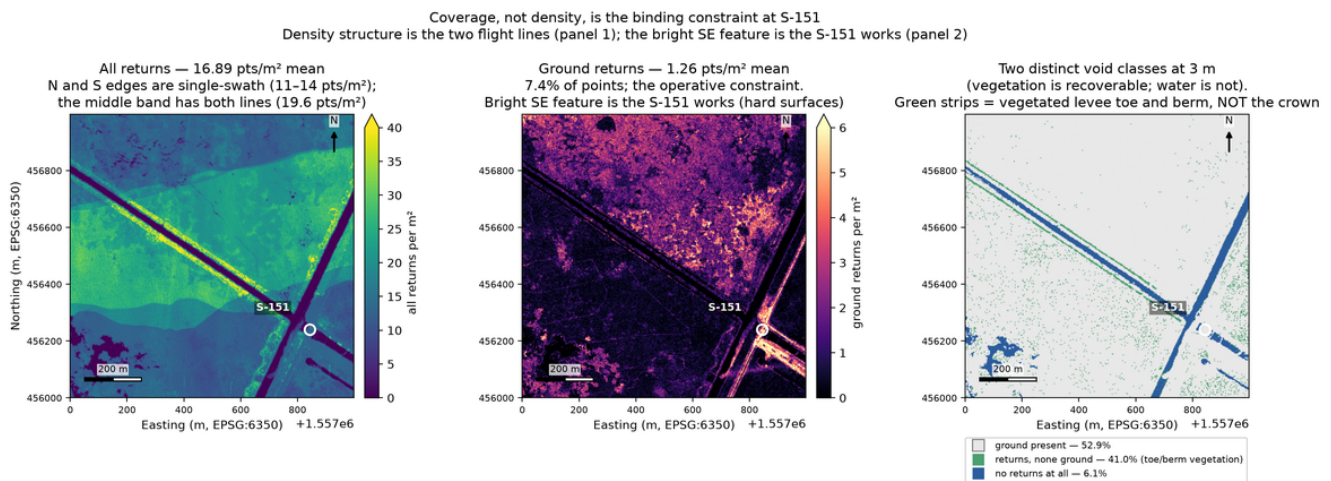


Figure 2. All returns, ground returns, and the two void classes. Density is high (16.89 pts/m²) while ground coverage is not (1.26 pts/m²). The density banding in panel 1 is the two flight lines (§7.5); the bright feature in panel 2 is the S-151 works.

4. Cell size, derived from coverage

Cell size was measured rather than carried over. The sweep rasterizes once at 0.5 m and aggregates by integer factors, so per-cell counts are exact and grid phase is identical at every size.

cell	no ground return	no returns at all	median ground pts/cell
0.5 m	76.79%	7.24%	0
1.0 m	47.04%	6.06%	1
2.0 m	18.11%	5.29%	3
3.0 m	8.29%	4.72%	8
5.0 m	4.63%	3.79%	22
10.0 m	2.21%	1.98%	89

The two columns behave differently, and that difference is the argument. Cells with no ground return fall steeply — 76.79% to 8.29% — because larger cells catch the sparse ground observations that penetrate the sawgrass. Cells with no returns at all barely move, from 7.24% to 4.72%, because open water does not return signal at any resolution. By 5 m the two columns nearly converge, meaning almost every remaining empty cell is genuine water

rather than missing ground.

3.0 m is taken as the working value — the fine end of the usable range, keeping ground-empty cells under 10% while preserving as much detail as the coverage supports.

For contrast, the prior desert project used a 3.3 ft (≈ 1.0 m) cell. Here that would leave **47.04%** of cells with no ground observation. This is the clearest single case in the project for deriving a parameter rather than inheriting it.

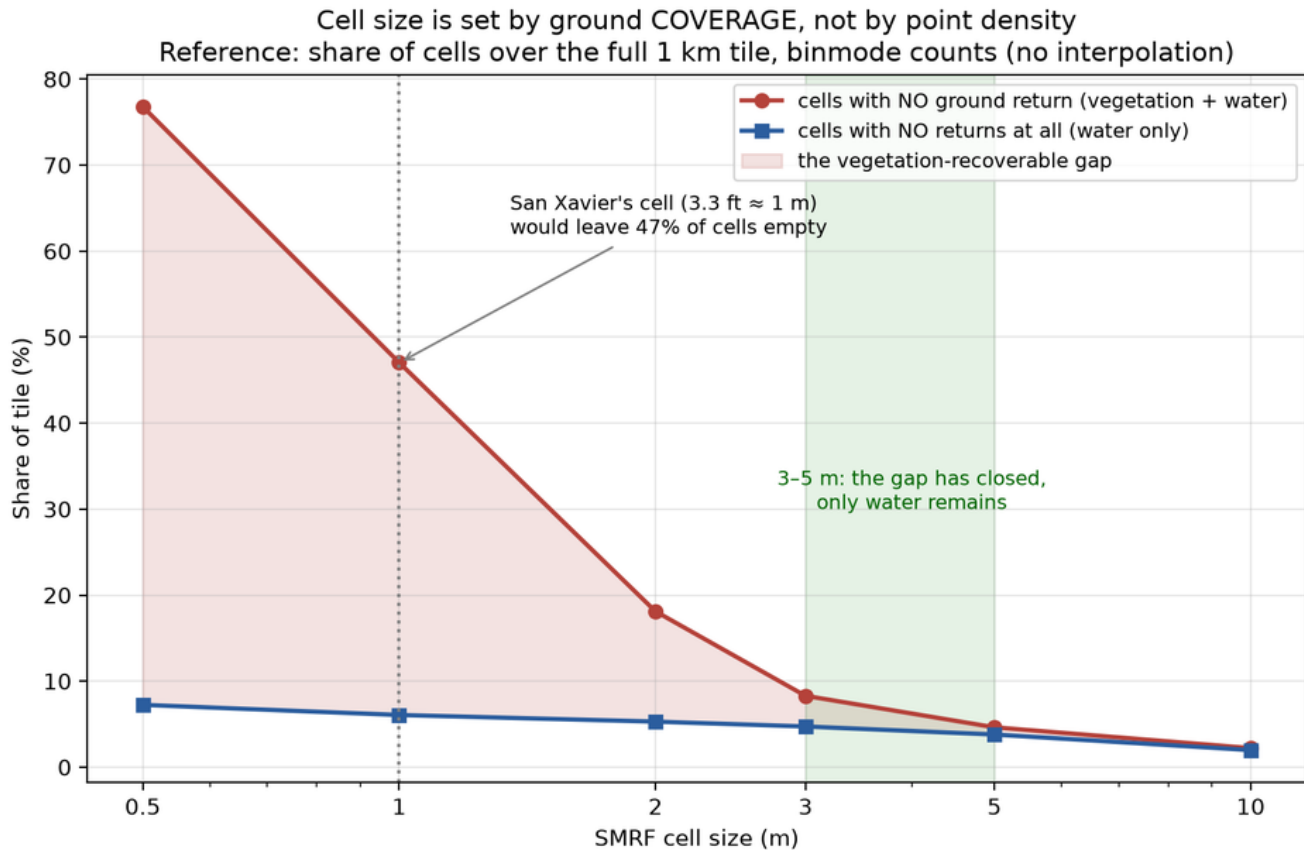


Figure 3. Cell-size sweep. Cells lacking a ground return fall steeply with cell size; cells lacking any return barely move, because open water returns nothing at any resolution. The gap between the curves is what larger cells recover.

5. Method and parameters

PDAL 2.10.0. Vendor classification is discarded and re-derived, so the result is independent of the vendor's decisions:

```
` readers.las -> filters.assign Classification[:] = 0 (strip vendor classes) -> filters.elm cell 10.0 m, threshold 1.0 m -> filters.outlier statistical, mean_k 8, multiplier 3.0 -> filters.smrf cell 3.0 m, window 3.0 m, slope 0.05, threshold 0.15 m, scalar 1.25 -> filters.range Classification[2:2] -> writers.gdal IDW, 3.0 m, fixed grid 333x333 @ 1557000/456000`
```

Five of the eight numeric parameters are measured from this tile; one is inherited without independent validation and is flagged as such. Full derivation, including a retracted first attempt at window, is in `parameter_derivation.md`. In summary:

parameter	value	basis
cell	3.0 m	measured — coverage sweep (§4)
window	3.0 m	measured — crown width vs. structuring element
slope	0.05	measured — marsh slope vs. vegetation height
threshold	0.15 m	measured — ground residual σ vs. vegetation height
scalar	1.25	inherited, not validated
ELM cell / threshold	10.0 m / 1.0 m	reasoned; low-impact stage

Every run writes its pipeline JSON to `scripts/pipelines/` as an audit trail, and outputs are compared by checksum rather than by eye — two parameter settings past SMRF's convergence point produce byte-identical rasters, which neither summary statistics nor a hillshade will reveal.

6. Agreement with the vendor surface

`dem_w3_s0.05_t0.15.tif` minus `dem_VENDOR_3m_aligned.tif`, both on an identical 333×333 grid at 3 m. 100% of cells valid in both.

The tile is not homogeneous, and a pooled statistic blends three populations that answer different questions. Regions are disjoint and cover the tile; "water" is cells with **zero returns of any kind** (binmode count, not a radius search).

region	cells	share	mean	median	std	RMSE	d >0.15 m
pooled	110,889	100%	+0.032	+0.038	0.106	0.111	2.16%
marsh	104,947	94.6%	+0.038	+0.039	0.072	0.081	1.79%
crest (levee top)	706	0.6%	−0.710	−0.911	0.651	0.963	59.9%
water	5,236	4.7%	+0.027	+0.028	0.063	0.069	1.74%

The marsh row is the meaningful one. Across 94.6% of the tile the two independent classifications agree to **0.081 m RMSE**, with a small positive bias (+0.038 m) — this surface sits marginally above the vendor's, consistent with a slightly more permissive ground call.

The pooled row should not be quoted. It averages real ground agreement together with a known truncation and with two interpolations across ground nobody measured.

The water row measures nothing about terrain. Neither surface observed the canal bed; both interpolate across it. That the two interpolations agree to 0.069 m says only that they were computed similarly. The delivered surface must not be read as bathymetry, and the canal surface is interpolated at a median **2.374 m** (cells with zero returns at 3 m, sampled on the aligned vendor surface) where there is in fact open water.

Correction, 2026-08-11: this figure previously read "roughly 2.28 m". That value does not reproduce under any tested definition -- median 2.374, mean 2.365, largest-water-body 2.381, p25 2.344 -- and its method was never recorded, so it cannot be checked. It is replaced by a value whose definition is stated. Same failure as the predecessor project's CHM cluster count, which also had no recorded search radius and also did not reproduce.

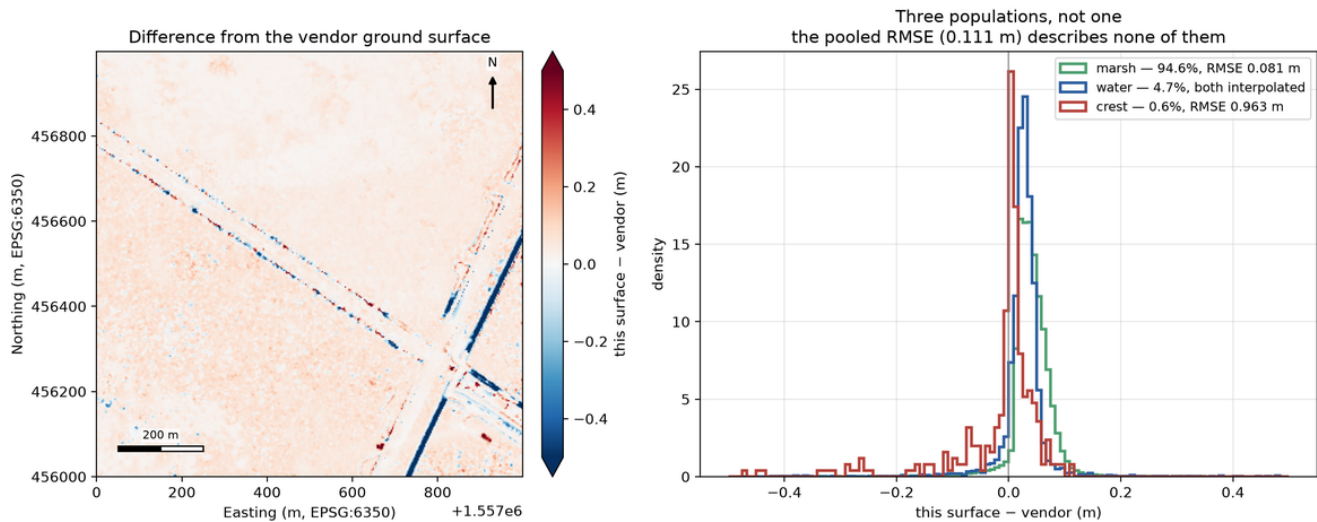


Figure 4. Difference from the vendor ground surface, and the three populations it contains. The marsh distribution (0.081 m RMSE) is the meaningful one; the pooled statistic describes none of the three.

7. Limitations

7.1 The L-67A crown is truncated by ~0.76 m — accepted and documented

The delivered surface truncates the L-67A embankment crown. Crest cells sit a median **-0.911 m** (mean -0.710 m) relative to the vendor surface; equivalently, median crest height above the marsh datum falls from **2.369 m to 1.612 m, a 0.757 m reduction**. This affects 706 cells, 0.64% of the tile.

The measured reason. SMRF removes a raised feature by morphological opening once its diamond structuring element no longer fits inside the feature *at the height being cut*. A levee is a wedge, so the governing width is the crown, not the base:

height above marsh	width
0.30 m	45.7 m
1.00 m	32.3 m
2.00 m	18.0 m
2.50 m	8.5 m
2.75 m	6.0 m

At `cell = 3.0 m` the smallest structuring element SMRF can form (radius 1) already spans **9 m**, which exceeds the crown above roughly 2.4 m. `window` is therefore already at its floor — any value $\leq \text{cell}$ yields radius 1 — and **no window setting at this cell size preserves the crown**. This is a resolution floor, not a tuning failure.

Both competing requirements, quantified. They are irreconcilable within one parameter set:

requirement	needs	measured basis
ground coverage in marsh	cell ≥ 3 m	at 1 m, 47.04% of cells contain no ground return; at 3 m, 8.31%
resolving the 6–8 m crown	cell ≤ 1.5 m	crown is 2–3 px at 3 m, narrower than the minimum structuring element

The finer cell is not a free improvement — this was tested. Cell 1.5 m does preserve the crown (99.6% retained, crest 1.989 m above marsh). But measured over marsh cells only, with the embankment and a 30 m buffer excluded so the feature under test cannot contaminate its own control, it also under-filters vegetation:

	marsh mean bias	marsh RMSE	marsh cells >0.15 m above vendor
cell 3.0 m (delivered)	+0.0416 m	0.0497 m	0.15%
cell 1.5 m	+0.0637 m	0.0760 m	2.34%

A **15× increase** in marsh cells sitting above the vendor surface (2.34% vs 0.15%), and a 53% higher marsh RMSE, across the 94.6% of the tile that is the actual deliverable.

Correction, 2026-08-11: this read "23×" until the deliverable-number audit flagged it [cited: superseded value, retained to show the error]. 23× is $2.34\% \div 0.10\%$, but 0.10% is the tail for w_6-w_{50} ; the delivered configuration is w_3 , whose tail is 0.15%. The comparison was against the wrong baseline — the same failure class as the retracted embankment-width derivation in `parameter_derivation.md`. The conclusion is unchanged; only its magnitude was overstated. The crown is bought by degrading everything else.

Decision: accept the truncation. The alternative — a dual-cell composite, fine near the levee and coarse in the marsh — would trade a measured, bounded, documented limitation for an undocumented seam along a boundary that would itself need justifying. Tile-edge discontinuity of exactly that kind was a real and quantified defect in the prior project. A known 0.76 m truncation on 0.64% of the tile is the better outcome, and it is stated here rather than discovered downstream.

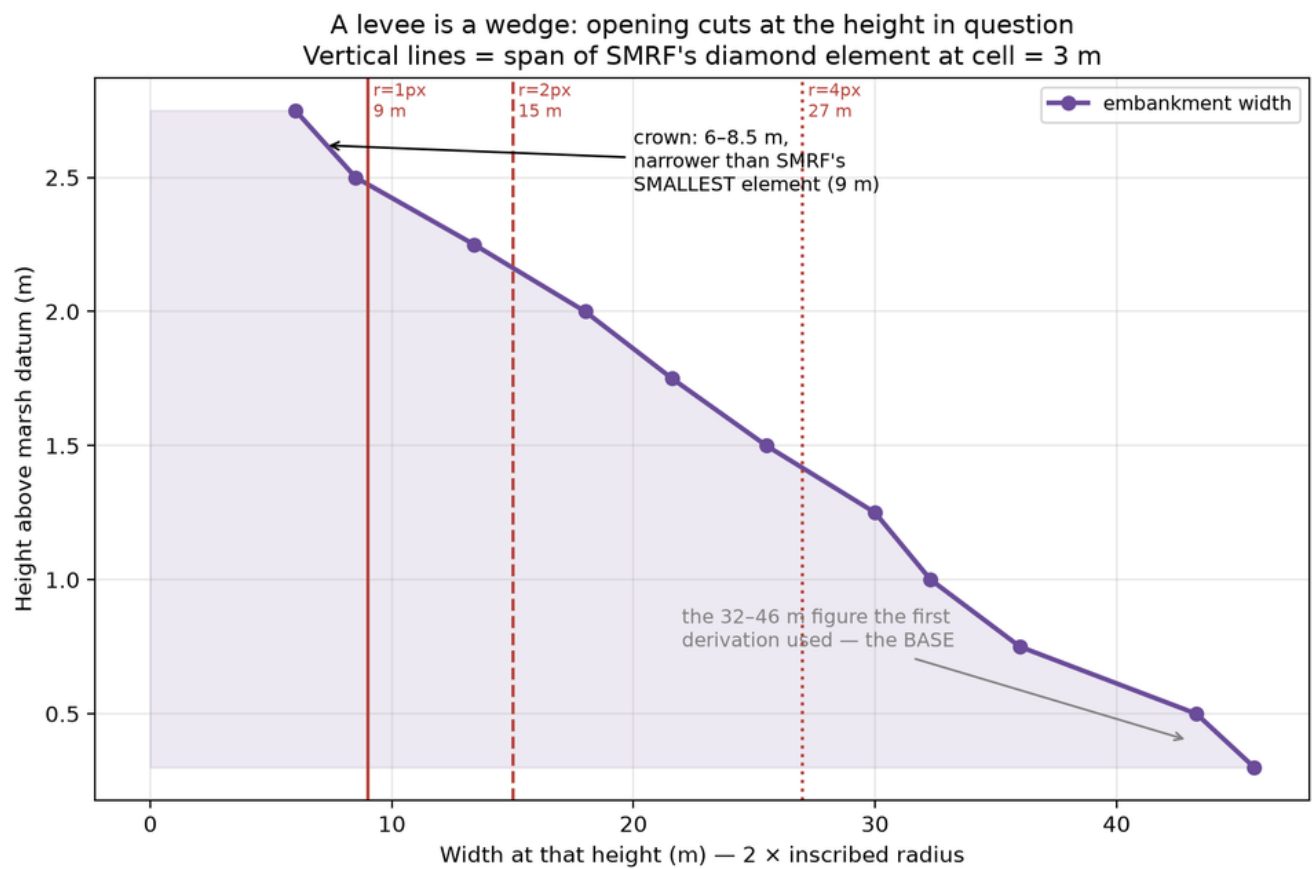


Figure 5. Embankment width against height above marsh. A levee is a wedge, and morphological opening acts on the width at the height being cut — so the 6–8.5 m crown governs, not the 32–46 m base. Vertical lines mark the span of SMRF's structuring element at cell = 3 m.

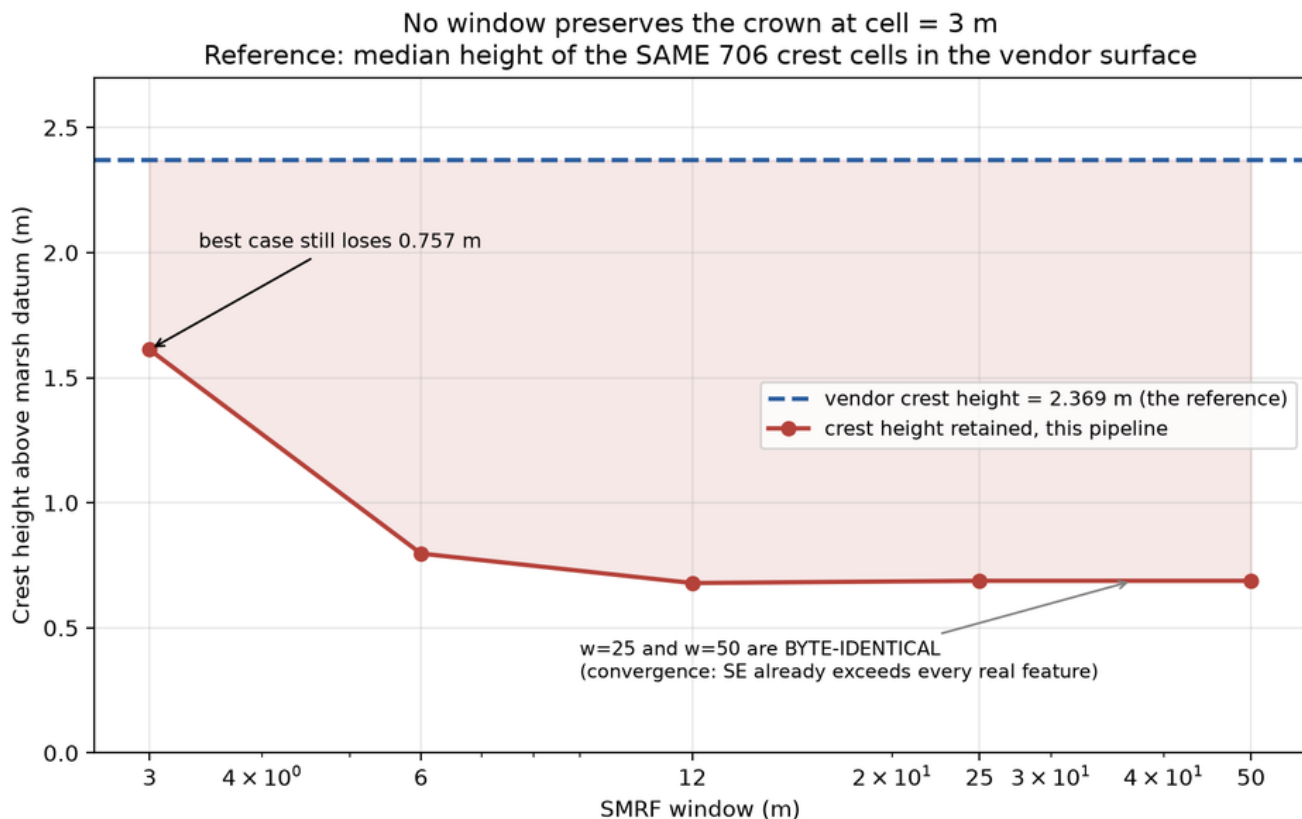


Figure 6. Crest height retained against `window`, referenced to the same 706 crest cells measured in the vendor surface. No setting preserves the crown; `window` = 25 m and 50 m produce byte-identical rasters, confirming convergence. The first derivation predicted survival below 12 m and was refuted — see the Appendix.

7.2 Bank flanks are classified non-ground

Terrain slope on this tile is **bimodal**: median 0.489%, p90 2.334%, but p99 **20.372%** and max 50.796% — a flat marsh plane plus engineered banks. Measured by `gdaldem slope -p` on the vendor 3 m surface **as originally gridded** (335x334, point-derived extent). On the aligned 333x333 grid the same measurement gives 0.490 / 2.318 / 20.442 / 51.088: the difference is grid phase, not disagreement, and is the same effect quantified in §7.5. `slope = 0.05` is set from the marsh, because setting it from the banks would put the radius-1 cutoff at 0.6 m, above the 90th percentile of vegetation height, and almost nothing would be removed. The cost is that steep bank flanks exceed the cutoff and are rejected. One parameter cannot serve both populations.

7.3 Ground and vegetation overlap irreducibly

Class-2 ground residual about the 3 m surface has std **0.072 m** (p5–p95 spread 0.133 m). Unclassified returns sit at p25 +0.050 m and p50 +0.240 m — but **40.6% fall below 0.15 m** above ground, 56.1% below 0.30 m, 70.3% below 0.50 m. The two populations genuinely overlap and no elevation threshold separates them cleanly. This is a property of sawgrass in a metre-relief marsh, not a parameter left untuned. `threshold = 0.15 m` is placed at roughly 2σ of genuine ground scatter, deliberately trading one error against the other.

7.4 Open water is absence, not error

6.07% of the tile returns nothing at all; 95.7% of that area lies in clusters larger than 100 m², i.e. the canals rather than scattered dropout. This is water absorbing the 1064 nm pulse [cited: sensor class specification] — irreducible, not a classification problem. See §6 on why the interpolated canal surface must not be presented as measured terrain.

7.5 Acquisition: two flight lines, flown 19.8 hours apart

The tile is covered by **two** flight lines — `PointSourceId` 34012 (north) and 34101 (south) — with GPS times 71,229 s apart, so the passes are on different days rather than one continuous lift. Coverage at 10 m is **48.8% single-swath, 49.2% double, 2.0% uncovered**.

The coverage is banded, not uniform, and a downstream user will meet this as an unexplained density step:

band	northing	coverage	all-return density
north edge	> 456,892	swath 34012 only	12.5–13.6 pts/m ²
middle	456,237–456,892	both swaths	19.6 pts/m²
south edge	< 456,237	swath 34101 only	10.6–11.6 pts/m ²

The step across northing 456,237 is roughly 2×, runs the full tile width, and is acquisition geometry rather than terrain. Void behaviour follows it: 5.23% of single-swath cells receive no returns at 1 m against 3.11% of double-swath cells. Anyone mosaicking this tile against a neighbour, or resampling it, should expect that discontinuity and not read it as a property of the marsh.

Measured inter-swath offset: −0.0027 m (marsh, 21,723 cells observed by both), RMSE 0.0287 m. Method: vendor class-2 points split by `PointSourceId`, each rasterized on the identical explicit grid with no IDW fallback, compared only where both swaths actually observed ground. SMRF was deliberately not re-run per swath — it anchors its internal grid to the input extent, and two swaths have different extents (§5.6).

An overnight gap was expected to produce a *larger* offset than the predecessor project's +0.124 ft (37.8 mm) single-day figure [cited: lidar-portfolio CLAUDE.md], since GNSS conditions and wetland water level both change overnight. It did not: 2.7 mm, an order of magnitude smaller.

This bears on §6. The 0.0287 m inter-swath RMS is present in the vendor surface and this one alike, since both are built from all returns regardless of swath. It is therefore a component of the 0.081 m marsh agreement figure that is **not** classification disagreement — roughly 13% of its variance. The remainder is genuinely classification.

Coverage is not stratified by acquisition. Single-swath cells are 44.24% no-ground and double-swath cells 47.69%, against the tile-wide 47.04% — a small spread running *opposite* to "degraded single coverage". Dropout behaves as expected: 5.23% no-returns at one swath versus 3.11% at two. So the cell-size sweep in §5.1 is not an average across good and bad acquisition.

7.6 Single tile, no adjacent context

Classification near the tile edge lacks neighbourhood context from outside the tile, and no buffering from adjacent tiles was implemented here. The effect is bounded by SMRF's reach — $\text{at window} = 3 \text{ m, } \text{ceil}(\text{window}/\text{cell}) = 1 \text{ cell}$ — so it is far smaller than in the prior project, where a 122 ft reach [cited: lidar-portfolio CLAUDE.md] produced measurable seam discontinuity. Not measured on this tile; noted as unquantified.

8. Deliverables

file	description
output/dem/dem_w3_s0.05_t0.15.tif	bare-earth DEM, 3 m, IDW (md5 ab0bebafe...)
output/dem/dem_VENDOR_3m_aligned.tif	vendor class-2 reference, identical grid
output/dem/count_allret_3m_aligned.tif	all-return count, binmode (water mask)
output/reports/parameter_derivation.md	full derivation incl. retracted attempt
output/reports/vendor_baseline_characterization.md	coverage / void analysis
scripts/pipelines/*.json	one pipeline per run, audit trail

Reproduce with `scripts/run_smrf.py`; re-derive every table above with `measure_window_sweep.py`, `measure_embankment_profile.py`, `check_veg_cost.py` and `qc_vs_vendor.py`. Each states its measurement parameters in its own docstring.

Appendix A. Parameter derivation in full

Tile: USGS_LPC_FL_Southeast_2018_D18_SUPPLEMENTAL_e1557n0456.laz **CRS:** EPSG:6350, NAD83(2011) Conus Albers — **metres** **Date:** 2026-08-11

Every parameter below is labelled **MEASURED** (derived from this tile's own data, with the measurement stated) or **INHERITED** (carried from project one without independent validation here). Nothing is scaled proportionally from San Xavier's values; that shortcut is what produced a retracted finding in project one.

1. The mechanism these parameters act through

Read from PDAL 2.10.0 source (`filters/SMRFilter.cpp`, `progressiveFilter()` and `classifyGround()`), not from CLI help text:

```
` max_radius = ceil(window / cell) # in PIXELS for radius in 1 .. max_radius: erode by 1 (cumulative), then dilate by
radius # diamond SE threshold = slope cell radius # progressive cutoff ... thresh = threshold + scalar *
surface_slope # final classification `
```

Two consequences drive everything below.

Opening removes an elevated feature once the structuring element no longer fits inside it. A diamond SE of radius r pixels fits within a feature of width W pixels while $w \geq 2r + 1$; the feature is eroded away once $2r + 1 > w$. So a feature of width W survives all windows below roughly $w / 2$ and is destroyed above it.

window therefore converges. Once `max_radius` exceeds the largest real non-ground feature present, further increases change nothing and produce byte-identical output. A sweep that does not straddle a real feature scale tests nothing — project one recorded a `window` finding twice on this exact mistake, and both were retracted.

2. `cell = 3.0 m` — **MEASURED**

From the vendor-baseline coverage sweep (see `vendor_baseline_characterization.md`): share of cells containing no ground return, by cell size.

cell	no ground point	no returns at all
0.5 m	76.79%	7.24%
1.0 m	47.04%	6.06%
2.0 m	18.11%	5.29%
3.0 m	8.31%	4.72%
5.0 m	4.63%	3.79%
10.0 m	2.21%	1.98%

The vegetation-driven gap closes by 3–5 m while the water floor does not; by 5 m the two curves nearly converge, meaning almost every remaining empty cell is genuine open water. 3.0 m is taken as the working value — the low end of the measured range, preserving the most detail while keeping ground-empty cells under 10%.

San Xavier's 3.3 ft (≈ 1.0 m) would leave **47%** of cells with no ground point here. This is the clearest single case for deriving rather than carrying over.

3. _{window} = the question the sweep answers — MEASURED

Two feature scales matter, and they are measured, not assumed:

feature	measured width	eroded once window exceeds
vegetation clumps	~ 6.3 m (2×3.16 m p99 distance-to-ground)	~ 3 m
L-67A embankment	32–46 m (crest 2.91 m above marsh)	~ 16–23 m

The scales are separated by roughly 5 $\times$, so a window exists that erodes vegetation while never reaching embankment scale.

The embankment must be KEPT. It is a constructed levee — earth, not vegetation or structure — and a bare-earth DEM of this site that omits it is wrong. This is a design decision, stated rather than buried: it is the reason window is bounded above at all.

At `cell = 3.0 m`, `max_radius = ceil(window / 3)` pixels and the SE spans `2r + 1` pixels:

window	max_radius	SE span	vs. 32–46 m embankment
6 m	2 px	15 m	well inside — crest survives
12 m	4 px	27 m	inside — crest survives
25 m	9 px	57 m	exceeds — crest destroyed
50 m	17 px	105 m	far exceeds — crest destroyed

Prediction, recorded before the runs: 6 m and 12 m preserve the embankment; 25 m and 50 m destroy it; 25 m and 50 m may be byte-identical to each other if both sit past convergence.

RESULT: the prediction was **WRONG**, and the error was the input, not the arithmetic

Measured with `scripts/measure_window_sweep.py` (crest mask fixed from the aligned vendor raster; parameters in that script's docstring):

window	crest above marsh	crest kept	marsh drift	md5
vendor	2.369 m	—	—	—
6 m	0.797 m	41.1%	+0.034 m	70b1f8b4...
12 m	0.679 m	30.3%	+0.034 m	4a3f0eb0...
25 m	0.688 m	29.6%	+0.034 m	576d7559...
50 m	0.688 m	29.6%	+0.034 m	576d7559...

The embankment is **already destroyed at the smallest window tested**. Window moves the crest by 0.12 m across an 8× range while the vendor surface sits 2.37 m above marsh. The marsh control is flat at +0.034 m throughout, so this is the crest specifically and not a global shift.

Two parts of the prediction *did* hold, which is precisely why the headline being wrong matters: 25 m and 50 m are **byte-identical**, so convergence sits between 12 and 25 m exactly where `max_radius = ceil(window/cell)` says it should. The mechanism was read correctly. It was applied to the wrong number.

The error: the recorded 32–46 m width was measured by thresholding 0.3–1.5 m above marsh — that is the embankment's **base**. Morphological opening acts on whatever width exists *at the height being cut*, and a levee is a wedge. Measured properly (`scripts/measure_embankment_profile.py`, inscribed-disk diameter so the diagonal orientation doesn't inflate it):

height above marsh	width	largest SE that fits
0.30 m	45.7 m	window ≤ 21 m
1.00 m	32.3 m	window ≤ 12 m
1.50 m	25.5 m	window ≤ 9 m
2.00 m	18.0 m	window ≤ 6 m
2.25 m	13.4 m	window ≤ 3 m
2.50 m	8.5 m	none at cell 3 m
2.75 m	6.0 m	none at cell 3 m

The crown is **6–13 m wide** — 2–4 px at `cell = 3 m`. SMRF's smallest structuring element (radius 1) already spans 3 px = 9 m, which exceeds the crown above ~2.4 m. **No window at cell = 3 m can preserve the top of this levee**, because the constraint binds at the resolution floor, not at the window.

This is the same class of error as project one's retracted seam finding: a correct mechanism, a real measurement, and a comparison against the wrong reference quantity. Recording the *parameters* of the original width measurement — "thresholded at 0.3–1.5 m" — is what made the error findable at all.

Re-derived from the crown, then tested

With the crown width in hand, two further runs were predicted **before** being run: `window = 3 m` at cell 3 m (`max_radius` 1 px, SE spans 9 m — holds while width ≥ 9 m, i.e. to ~2.45 m) and the same at cell 1.5 m (`max_radius` 2 px, SE spans 7.5 m — holds to ~2.55 m).

run	crest above marsh	crest kept	predicted	md5
w3, cell 3.0	1.612 m	72.4%	1.5–2.0 m ✓	ab0bebaf...
w6, cell 3.0	0.797 m	41.1%	—	70b1f8b4...
c1.5, w3	1.989 m	99.6%	1.8–2.2 m ✓	45d2c79a...

Both hit. **Which is exactly when this project's rule says to stop and check the mechanism instead of writing it up.**

The competing explanation, tested and half-confirmed

A smaller structuring element filters less *everywhere*. If that were the cause, the crown would survive not because the SE fits inside it but because the run retains more non-ground points tile-wide — vegetation included. The two explanations differ measurably: opening-scale effects are confined to features near SE scale, under-filtering is tile-wide.

Measured with `scripts/check_veg_cost.py` over marsh cells only, with the embankment plus a 30 m buffer excluded so the feature under test cannot contaminate its own control (77,987 cells, 70.3% of tile):

run	marsh mean bias	marsh RMSE	marsh >0.15 m high
w3	+0.0416 m	0.0497 m	0.15%
w6 / w12 / w25 / w50	+0.0414 m	0.0481 m	0.10%
c1.5, w3	+0.0637 m	0.0760 m	2.34%

For window: rejected. Across an 8× range the marsh statistics are identical to four decimals. The window effect is genuinely confined to structuring-element scale, so w3's better crest is not bought by retaining vegetation. The mechanism is verified independently of the prediction that proposed it.

For cell: confirmed. Cell 1.5 m shows a **15×** increase in marsh cells more than 0.15 m above the vendor surface (2.34% vs 0.15%) and a **53%** higher marsh RMSE (0.0760 vs 0.0497 m). Its 99.6% crown preservation is substantially bought by retaining non-ground returns. Its crest number alone would have recommended it.

Correction, 2026-08-11 [cited: superseded values, retained to show the error]: this section previously read "23×" and "58%". Both compared cell 1.5 m against the w6–w50 figures (0.10% tail, 0.0481 m RMSE) rather than against the **delivered** w3 run (0.15%, 0.0497 m). Same wrong-reference error as the retracted window derivation above, and it reached two numbers rather than one. The memo was corrected first and this file was missed; the deliverable-number audit caught the disagreement between them. Conclusion unchanged, both magnitudes were overstated.

3b. $\text{window} = 3.0 \text{ m}$ — MEASURED, with a stated residual limitation

At $\text{cell} = 3.0 \text{ m}$, $\text{max_radius} = \text{ceil}(\text{window} / \text{cell})$, so **any window ≤ 3 m yields radius 1** — the minimum SMRF can use. The site's feature scales drive this parameter to its floor.

Residual limitation, stated rather than buried: even at the minimum, the recovered crest is 1.612 m above marsh against the vendor's 2.369 m. **The top ~0.76 m of the levee is not recoverable at this cell size**, because the crown above ~2.4 m is narrower (6.0–8.5 m) than the smallest structuring element SMRF can form (9 m). This is a resolution floor, not a tuning failure, and no window value addresses it.

The available remedies both cost more than they return here: a finer cell resolves the crown but retains marsh vegetation (measured above), and a region-specific parameter set would mean two surfaces mosaicked along a boundary that itself has to be justified. Neither is taken; the truncation is documented instead.

4. $\text{slope} = 0.05$ — MEASURED

Terrain slope from the vendor ground surface at 3 m (`gdaldem slope -p`):

percentile	slope
median	0.489%
p75	1.10%
p90	2.33%
p95	4.96%
p99	20.4%
max	50.8%

The distribution is **bimodal**: a nearly flat marsh plane plus engineered banks. One value must serve both, and that tension is real rather than a tuning nuisance.

$\text{slope} = 0.05$ is set from the marsh, not the banks. At radius 1 the progressive cutoff is $0.05 \times 3 \times 1 = 0.15$ m, which sits just above the measured class-2 residual scatter (std 0.072 m, p5–p95 spread 0.133 m) and just below the median vegetation height above ground (+0.240 m) — so it rejects typical vegetation without flagging real ground noise. Setting it from the p99 bank slope instead (0.2) would put the radius-1 cutoff at 0.6 m, above the *90th percentile* of vegetation height, and almost nothing would be removed.

Known cost, not hidden: at this value the steep bank flanks exceed the cutoff and will be flagged non-ground. The crest is protected by `window`, not by `slope`. Flank treatment is a genuine limitation of a single-parameter fit to a bimodal surface.

5. `threshold = 0.15 m` — MEASURED

The final classification base threshold, from the measured overlap between the two populations:

- class-2 ground residual about the 3 m surface: **std 0.072 m**, p5–p95 spread 0.133 m
- unclassified returns above ground: p25 **+0.050 m**, p50 **+0.240 m**, with **40.6% below 0.15 m**, 56.1% below 0.30 m, 70.3% below 0.50 m

$0.15 \text{ m} \approx 2\sigma$ of genuine ground scatter, so it accepts real ground while sitting below the median vegetation return.

The populations genuinely overlap and no threshold separates them. 40.6% of unclassified returns sit within 0.15 m of ground — that is a property of sawgrass in a metre-scale-relief marsh, not a parameter yet to be tuned. Any value here trades one error against the other; this one is placed at the measured ground-noise ceiling deliberately.

6. `scalar = 1.25` — INHERITED, NOT VALIDATED

Carried from project one, where it was itself never independently validated. It scales the final threshold by local surface slope ($\text{thresh} = \text{threshold} + \text{scalar} \times \text{surface_slope}$), so on this tile it mostly matters on the bank flanks — the one place the parameter set is already known to be weakest.

Stated as inherited rather than presented as derived. Testing it is a reasonable follow-on and has not been done.

7. `ELMcell = 10.0 m`, `threshold = 1.0 m` — REASONED, low stakes

ELM removes low outliers before classification. This tile carries 468 class-7 low-noise points (0.003%), so the stage is low-impact by construction.

10 m / 1.0 m are the metric equivalents of PDAL's own defaults, used deliberately here rather than converted from project one's feet values — those were the literals that turned out to be hardcoded ($33.0 / 3.3$ ft) inside a function

labelled generic, which would have become a 33 m cell and 3.3 m threshold against this CRS.

1.0 m is kept conservative against 2.81 m of total relief: the canal banks drop 2+ m over short distances and a tighter threshold risks eating real bank returns.

8. Summary

Working parameter set — `dem_w3_s0.05_t0.15.tif`, md5 `ab0beba...`:

parameter	value	basis
cell	3.0 m	MEASURED — coverage sweep (§2)
window	3.0 m	MEASURED — crown width vs. SE span (§3, §3b)
slope	0.05	MEASURED — marsh slope vs. vegetation height (§4)
threshold	0.15 m	MEASURED — ground residual σ vs. vegetation height (§5)
scalar	1.25	INHERITED — not validated (§6)
ELM cell	10.0 m	reasoned (PDAL metric default)
ELM threshold	1.0 m	reasoned, conservative vs. bank relief
output res	3.0 m	matches cell

Five of eight are measured from this tile. One is explicitly inherited without validation. Two are reasoned defaults on a low-impact stage.

Known limitations of this set, none of them tunable away:

1. **The levee crown is truncated ~0.76 m** (§3b). The crown is narrower than SMRF's smallest structuring element at this cell size.
2. **Bank flanks are classified non-ground** (§4). `slope = 0.05` is set from the marsh; the surface is bimodal and one value cannot serve both populations.
3. **Ground and vegetation overlap irreducibly** (§5). 40.6% of unclassified returns sit within 0.15 m of ground — sawgrass, not a threshold yet to be found.
4. **Open water is absence, not error** (§2). 6.07% of the tile returns nothing; the surface interpolates across the canal and must not be presented as measured terrain there.

Reproduce any run with `scripts/run_smrf.py`; every run writes its pipeline JSON to `scripts/pipelines/`. Re-derive the tables above with `measure_window_sweep.py`, `measure_embankment_profile.py` and `check_veg_cost.py` — each states its own parameters in its docstring.